EXERCISES List 3 October 21, 2026

Simulations and algorithmic applications of Markov chains

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Coupling

L3.1. Let $M \geq 2$ and $p \in (0,1)$. Consider a Markov chain on $\mathbb{E} = \{1, \ldots, M\}$ with the following update rule. From state i,

- with probability $1-p$, choose the next state uniformly from $\mathbb{E}$;
- with probability p, move to $i+1$ if $i < M$, and to 1 if $i = M$.

Thus the transition matrix is

$$\mathbf{P}(i,j) = \begin{cases} p + \dfrac{1-p}{M}, & \text{if } (i < M,\ j = i+1) \text{ or } (i = M,\ j = 1), \\ \dfrac{1-p}{M}, & \text{otherwise.} \end{cases}$$

All random choices at a new step are independent of the past.

(a) Is the chain ergodic? Justify your answer.

(b) Find its stationary distribution $\boldsymbol{\pi}$.

(c) Construct a coupling $(\mathbf{X}, \mathbf{Y})$ of two chains with this transition matrix, starting at arbitrary states $X_0 = i$, $Y_0 = j$. Check that both marginal transition laws are correct, and arrange that once the chains meet, they stay together. For your coupling, define $T = \inf\{k \geq 0 : X_k = Y_k\}$. Calculate or bound $\mathbb{P}(T > k)$, and give an upper bound tending to zero as $k \to \infty$ that is valid uniformly over the initial states i, j.

L3.2. Let $M \geq 2$ and $p \in (0,1)$. Consider a Markov chain on $\mathbb{E} = \{1, \ldots, M\}$ with the following update rule. From state i,

- with probability $1-p$, move to 1;
- with probability p, move to $\min(i+1, M)$.

Thus at state M, the second option leaves the state unchanged. Thus the transition matrix is

$$\mathbf{P}(i,j) = \begin{cases} 1-p, & j = 1, \\ p, & j = \min(i+1, M), \\ 0, & \text{otherwise.} \end{cases}$$

All random choices at a new step are independent of the past. Answer the same questions (a), (b), and (c) as in Exercise L3.1.

L3.3. Let $\mathbb{E} = \{1, \ldots, M\}$, where $M \geq 3$. Consider the simple random walk on the complete graph without loops: from state i, choose the next state uniformly among all other states. Thus

$$\mathbf{P}(i,j) = \begin{cases} 1/(M-1), & j \neq i, \\ 0, & j = i. \end{cases}$$

Answer the same questions (a), (b), and (c) as in Exercise L3.1. In (c), make the probability of meeting at the next step as large as possible whenever the states differ.

(d) What changes when $M = 2$?

L3.4. Revisit Exercise L2.17 from List 2, using the coupling constructed with a common update function. Its meeting time T^{dep} has a geometric distribution, so its tail can be calculated exactly. For $\alpha = 3/4$ and $\beta = 1/2$, instead of calculating $\mathbb{P}(T^{\text{dep}} > k)$ directly, compare the bounds obtained using

(a) Markov's inequality;

(b) Chebyshev's inequality.

Reversibility

Recall. A Markov chain with transition matrix $\mathbf{P}$ on $\mathbb{E}$ is *reversible* if there exists a probability distribution $\boldsymbol{\pi}$ with $\pi(\mathbf{e}) > 0$ for every $\mathbf{e} \in \mathbb{E}$ such that

$$\pi(\mathbf{e})\mathbf{P}(\mathbf{e},\mathbf{e}') = \pi(\mathbf{e}')\mathbf{P}(\mathbf{e}',\mathbf{e}) \qquad \text{for all } \mathbf{e},\mathbf{e}' \in \mathbb{E}.$$

These are the *detailed balance equations*. Recall the lemma from the lecture: any probability distribution satisfying detailed balance is stationary, i.e., $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$.

L3.5. Let $\mathbf{X} = (X_0, X_1, \ldots)$ be a reversible Markov chain on $\mathbb{E}$ with transition matrix $\mathbf{P}$ and reversible distribution $\boldsymbol{\pi}$. Show that, if $X_0 \sim \boldsymbol{\pi}$, then for every $n \geq 1$ and every $\mathbf{e}_{i_0}, \ldots, \mathbf{e}_{i_n} \in \mathbb{E}$,

$$\mathbb{P}(X_0 = \mathbf{e}_{i_0}, X_1 = \mathbf{e}_{i_1}, \ldots, X_n = \mathbf{e}_{i_n}) = \mathbb{P}(X_0 = \mathbf{e}_{i_n}, X_1 = \mathbf{e}_{i_{n-1}}, \ldots, X_n = \mathbf{e}_{i_0}).$$

L3.6. Show that, if a transition matrix $\mathbf{P}$ is *doubly stochastic* (i.e., each column also sums to 1), then the uniform distribution is stationary.

L3.7. Let $\mathbf{P}$ be the transition matrix of an ergodic chain with stationary distribution $\boldsymbol{\pi}$. The transition matrix of the time-reversed chain is

$$\widetilde{\mathbf{P}}(\mathbf{e}_i, \mathbf{e}_j) = \frac{\pi(\mathbf{e}_j)}{\pi(\mathbf{e}_i)}\mathbf{P}(\mathbf{e}_j, \mathbf{e}_i).$$

Show that $\boldsymbol{\pi}$ is also stationary for $\widetilde{\mathbf{P}}$. The original chain need not be reversible; it is reversible precisely when $\widetilde{\mathbf{P}} = \mathbf{P}$.

L3.8. Let $G = (V, K)$ be a finite, simple (no loops), connected undirected graph. Write $\deg(v)$ for the degree of $v \in V$. Consider the simple random walk on G: from the current vertex v, choose the next vertex uniformly among its neighbors, so that

$$\mathbf{P}(v,w) = \begin{cases} 1/\deg(v), & \{v,w\} \in K, \\ 0, & \text{otherwise.} \end{cases}$$

Assuming that the chain is aperiodic, give a formula for its stationary distribution $\pi(v)$.

. Modify the rule by staying at the current vertex with probability $\alpha \in (0,1)$ and moving as before with probability $1-\alpha$. The new transition matrix is $\mathbf{P}' = \alpha\mathbf{I} + (1-\alpha)\mathbf{P}$. If $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$, then

$$\boldsymbol{\pi}\mathbf{P}' = \alpha\boldsymbol{\pi}\mathbf{I} + (1-\alpha)\boldsymbol{\pi}\mathbf{P} = \alpha\boldsymbol{\pi} + (1-\alpha)\boldsymbol{\pi} = \boldsymbol{\pi}.$$

Thus the modified chain has the same stationary distribution. This modification guarantees aperiodicity.

L3.9. A lonely chess king stands on a standard 8×8 board. At each step, it moves uniformly to one of the squares reachable by a legal king move.

(a) Is the resulting Markov chain aperiodic and/or irreducible?

(b) Find its stationary distribution.

L3.10. Let $\mathbf{P}$ be the transition matrix of an ergodic chain with stationary distribution $\boldsymbol{\pi}$. The chain need not be reversible. Define $\mathbf{M} = \mathbf{P}\widetilde{\mathbf{P}}$, where $\widetilde{\mathbf{P}}$ is the transition matrix of the time-reversed chain.

(a) Show that $\mathbf{M}$ is stochastic.

(b) Show that $\boldsymbol{\pi}$ is stationary for $\mathbf{M}$. Is the chain with transition matrix $\mathbf{M}$ reversible?